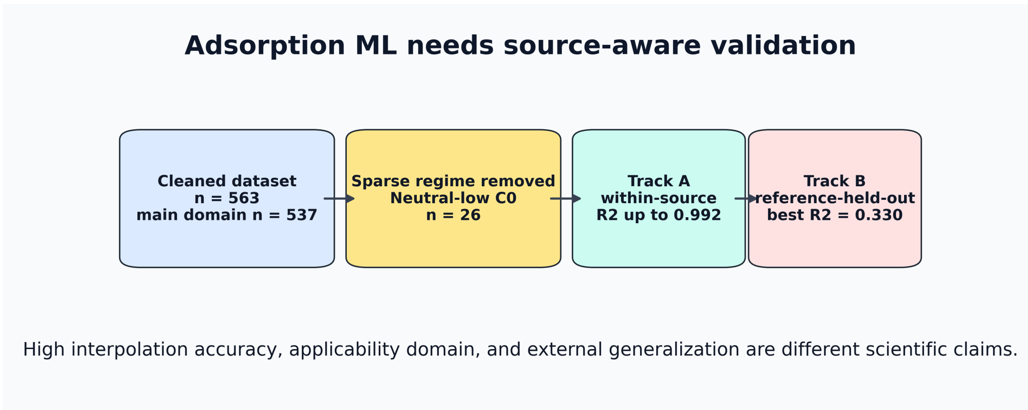
Graphical Abstract



Highlights

- Source-specific adsorption models reached R^2 up to 0.992.
- Sparse neutral-low-concentration cases were excluded from primary modeling.
- Row-wise validation overstated source-transfer performance.
- Reference-held-out sensitivity reached R^2 = 0.330 in the main domain.
- Capacity-inclusive and capacity-free models served distinct uses.

Source-aware machine learning for heavy-metal adsorption: distinguishing high-accuracy interpolation from cross-source generalization

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Abstract

Machine learning is increasingly used to predict adsorption removal efficiency for water and wastewater treatment, but reported accuracy can depend strongly on feature choice, applicability-domain coverage, and validation design. This study develops a source-aware and leakage-aware validation framework for heavy-metal adsorption removal-efficiency prediction using a literature-compiled experimental dataset. Source awareness was introduced by tracking the literature reference from which each adsorption record originated. Leakage awareness was addressed by testing whether model performance remained reliable when familiar rows, adsorbents, or entire literature sources were removed from training, and by separating feature sets that use adsorption capacity from feature sets designed for capacity-free pre-experiment screening. The raw dataset contained 566 adsorption records; after deterministic numeric parsing and quality filtering, 563 valid records remained. Because the neutral/low-concentration

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regime contained only 26 observations, primary modeling was restricted to a 537-record domain covering acidic/high-concentration, acidic/low-concentration, and neutral/high-concentration conditions. Four feature sets represented different levels of information availability: a capacity-inclusive process model and three capacity-free screening models based on ion descriptors, adsorbate identity, or numeric operating and material variables only. Track A quantified source-specific interpolation through within-source cross-validation, where XGBoost reached R^2 values of 0.992, 0.974, and 0.960 for three major source groups. Track B tested broader generalization using random row-wise, adsorbent-held-out, and reference-held-out splits. In the restricted primary domain, the full capacity-inclusive LightGBM model achieved $R^2 = 0.806$ under random row-wise validation. Under stricter validation, the best adsorbent-held-out performance was obtained by a capacity-free numeric LightGBM model ($R^2 = 0.532$), while reference-held-out performance reached $R^2 = 0.330$ for a capacity-inclusive LightGBM model and $R^2 = 0.305$ for the best capacity-free model. These results show that high adsorption-prediction accuracy is meaningful for interpolation within known sources, but broader deployment requires explicit applicability-domain definition, careful separation of feature-use cases, and validation against unseen adsorbents and unseen literature sources.
Keywords: adsorption, heavy metals, removal efficiency, machine learning, XGBoost, source-aware validation, wastewater treatment

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1. Introduction

Heavy-metal contamination remains a persistent challenge for water and wastewater treatment because metals such as Pb, Cd, Cr, Ni, Cu, As, Hg, and Zn are toxic, non-biodegradable, and prone to accumulation in environmental and biological systems. Adsorption is widely investigated for heavy-metal removal because it can operate under mild conditions, can be adapted to low-cost or waste-derived adsorbents, and can be optimized through pH, adsorbent dose, contact time, temperature, surface chemistry, and initial contaminant concentration. The expansion of experimental adsorption datasets has therefore encouraged the use of machine learning (ML) to predict removal efficiency and accelerate adsorbent screening.

Recent adsorption and environmental-remediation ML studies demonstrate that high predictive accuracy is possible. Hafsa et al. modeled heavy-metal adsorption efficiencies using multiple ML algorithms and reported R^2 values in the 0.96–0.99 range under ten-fold cross-validation, with experimental data supplemented by synthetic interpolation [1]. Lu et al. predicted heavy-metal removal by chitosan-based flocculants using random forests and reported $R^2 = 0.9354$, identifying solution pH and flocculant molecular weight as important drivers [2]. Wang et al. used interpretable ML for heavy-metal removal by biochar and reported XGBoost test performance near $R^2 = 0.99$, together with SHAP and partial-dependence analyses to relate removal efficiency to biochar and operating variables [3, 6]. Recent ensemble-learning work on biochar adsorption similarly reported strong XGBoost performance for heavy-metal adsorption efficiency prediction [4]. Within the scope of *Journal of Environmental Chemical Engineering*, interpretable XGBoost modeling has also been applied to heavy-metal removal efficiency in electrokinetic remediation, demonstrating the journal relevance of transparent ML workflows for environmental treatment systems [5].

The central challenge is that high validation accuracy does not necessarily imply reliable generalization to unseen laboratories, adsorbents, pollutant systems, or literature sources. In many adsorption ML studies, performance is estimated using random row-wise validation, where individual experimental records are randomly divided between training and testing. This can be useful for measuring interpolation within a compiled dataset, but it may also place closely related records from the same paper, adsorbent series, or operating campaign in both the training and test sets. As a result, the test set can appear independent while still containing familiar source-specific patterns. In ML-based science, such data leakage and validation mismatch can produce overoptimistic estimates of deployable performance [7]. Environmental ML best-practice guidance similarly emphasizes careful data splitting, leakage management, randomness assessment, and evaluation design [8]. These concerns are particularly important for adsorption datasets because different studies may use different adsorbent synthesis routes, contaminant species, concentration ranges, analytical protocols, and reporting conventions.

Feature definition can further blur the interpretation of reported accuracy. Adsorption capacity and removal efficiency are both derived from concentration changes in many batch adsorption experiments:

$$\mathrm{RE}(\%) = 100\frac{C_0 - C_e}{C_0}, \tag{1}$$

$$q_e = \frac{(C_0 - C_e)V}{m}, \tag{2}$$

where C_0 and C_e are the initial and equilibrium concentrations, V is the solution volume, and m is the adsorbent mass. Because both RE and q_e depend on the concentration change $C_0 - C_e$, adsorption capacity can carry strong information about removal efficiency. A model that uses capacity is therefore useful for process monitoring or post-experiment analysis, where capacity has already been measured or estimated. However, the same model should be treated cautiously for pre-experiment screening, because the capacity of a candidate adsorbent under a new condition is usually not known before the experiment is performed. Therefore, capacity-inclusive and capacity-free models answer different practical questions and should not be interpreted in the same way. Recent PFAS adsorption modeling has similarly shown that random-split performance can be much stronger than leave-one-compound-out performance, illustrating that interpolation and transfer to unseen chemical domains are distinct tasks [9].

This study addresses these issues by developing a source-aware and leakage-aware validation framework for heavy-metal adsorption removal-efficiency prediction across a broad literature-derived dataset spanning 189 normalized source references, 219 adsorbents, and 16 adsorbate classes. Here, source awareness means tracking the literature reference from which each adsorption record originated, rather than treating all rows as fully independent observations. Leakage awareness means testing whether model performance remains reliable when familiar information is removed from training, and separating feature sets that may be appropriate after an experiment from feature sets that are suitable before an experiment.

The framework separates two aspects that are often mixed in adsorption ML studies: what information is available to the model, and how independent the test data are from the training data. Four feature sets were constructed to represent different levels of information availability. A full capacity-inclusive feature set included adsorption capacity and capacity-derived predictors and was interpreted as a process-monitoring or post-experiment analysis model. Three capacity-free feature sets excluded adsorption capacity and were interpreted as pre-experiment screening models, because the capacity of a candidate adsorbent is usually not known before testing. Among the capacity-free models, one represented adsorbates using ion descriptors, one used normalized adsorbate identity as a categorical label, and one used only numeric operating, material, and engineered process variables as a strict screening baseline.

Two validation tracks were then used to separate interpolation from broader generalization. Track A evaluated source-specific interpolation by training and testing within individual literature sources. This asks whether high R^2 values comparable to those reported in the adsorption ML literature can be reproduced inside known source groups. Track B evaluated broader generalization using increasingly strict split designs. In random row-wise validation, individual records were randomly divided between training and testing. In adsorbent-held-out validation, selected adsorbents were kept completely unseen during training and used only for testing. In reference-held-out validation, all records from selected literature sources were kept unseen during training and used only for testing. The novelty of this study is therefore not simply a new regression model, but a source-aware and leakage-aware reporting framework that clarifies when high R^2 reflects valid interpolation within known experimental domains and when additional validation is needed before claiming deployment to new adsorbents, studies, or laboratories.

2. Materials and methods

2.1. Dataset and target variable

The dataset was compiled from reported heavy-metal adsorption experiments. Each record corresponds to one adsorption condition with a reported removal efficiency. The raw dataset contained 566 rows and 13 columns. After numeric parsing, missing-value filtering, and constraining removal efficiency to the physically meaningful range 0–100%, 563 records remained. The cleaned dataset included 189 normalized source references, 219 adsorbents, and 16 normalized adsorbate classes. The primary modeling domain excluded the sparse neutral/low-concentration regime, leaving 537 records for model training and testing.

The target variable was removal efficiency, RE (%). Input variables included adsorbent name, adsorbate identity, surface area, adsorption capacity, initial concentration, contact time, adsorbent dose, agitation speed, initial pH, temperature, and source reference. The source identifier, denoted `Ref_group`, was used as a source-origin proxy because explicit laboratory-origin labels were not available for all records. The pH-concentration regime label was not used as a model target or as a model input feature; it was created only for stratified splitting, diagnostic reporting, and applicability-domain definition.

2.2. Numeric cleaning and categorical normalization

Literature-derived adsorption tables often contain nonuniform numeric formatting. Numeric entries were cleaned using deterministic rules: decimal commas were converted to decimal points, uncertainty and approximate symbols were removed, multiple values in one cell were reduced to the first numeric value, and nonnumeric text was parsed using regular expressions. Rows with missing values in required numeric variables were removed. Adsorbate names were normalized to reduce duplicates such as elemental and valence-state aliases. Adsorbent names were grouped into broad families, including carbon-based, mineral/oxide, polymer, biomass, and composite/other categories.

The cleaned data were partitioned into four pH-concentration regimes using pH = 7 as the acidic/near-neutral boundary and the dataset median initial concentration, $C_0 = 50$ ppm, as the low/high concentration boundary (Table 1). The median C_0 is the middle value of the initial-concentration distribution and provides a data-driven threshold for separating lower- and higher-concentration experiments. These regimes were not used as prediction targets and were not passed to XGBoost, LightGBM, or ExtraTrees as predictor variables. Instead, the models predicted RE (%) from the original experimental and material variables, including numerical pH and numerical initial concentration. Regime labels were used only for stratified random splitting and diagnostic analysis, ensuring that train-test splits preserved chemically meaningful coverage across major operating regions. The acidic regimes dominated the dataset, whereas the neutral/low-C_0 regime contained only 26 observations. Because this regime was too small for stable split-wise evaluation, it was excluded from the primary modeling domain and treated as outside the present applicability domain.

Table 1: pH-concentration regime distribution in the cleaned dataset.

Regime	Count	Mean pH	Mean C_0 (ppm)	Mean RE (%)
Acidic_HighC0	246	4.51	190.27	70.30
Acidic_LowC0	231	5.12	11.04	74.06
Neutral_HighC0	60	7.65	166.51	68.05
Neutral_LowC0	26	7.47	20.24	81.41

Neutral_LowC0 was excluded from primary modeling because it contained only 26 observations; the other three regimes formed the 537-record primary domain.

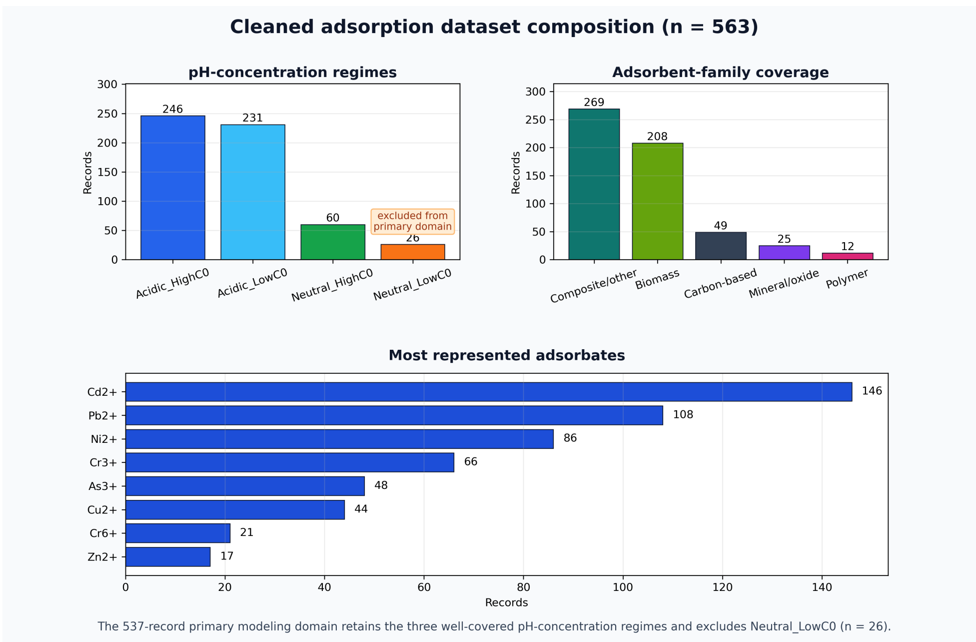

Figure 1: Composition of the cleaned adsorption dataset after deterministic numeric parsing and quality filtering. The sparse neutral/low-C_0 regime was retained in the dataset audit but excluded from the primary 537-record modeling domain.

2.3. Feature sets

Four feature sets were constructed to distinguish scientific use cases (Table 2). This separation is important because a model that includes adsorption capacity answers a different question from a model intended for pre-experiment adsorbent screening. The base experimental inputs were surface area, initial concentration, contact time, dose, agitation speed, initial pH, and temperature. The process-monitoring feature set additionally used adsorption capacity and capacity-derived terms, whereas the screening feature sets deliberately removed these terms.

Physics-inspired variables included site density, logarithmic contact time, mixing index, driving-force proxy, and acidity strength:

$$\text{Site Density} = \frac{\text{dose} \times \text{surface area}}{C_0}, \tag{3}$$

$$\text{LogTime} = \log(1 + \text{contact time}), \tag{4}$$

$$\text{Mixing Index} = \text{RPM} \times \text{LogTime}, \tag{5}$$

$$\text{Driving Force} = \frac{C_0}{\text{dose}}, \tag{6}$$

$$\text{Acidity Strength} = |\text{pH} - 7|. \tag{7}$$

Ion descriptors were added to represent adsorbates through simple aqueous-chemistry properties rather than only through pollutant-name labels. A one-hot adsorbate label identifies the metal species but does not encode chemical similarity between ions. Descriptors such as charge, ionic radius, hydrated radius, ionic potential, and hydration ratio allow the model to partially capture differences in ion size, charge density, and hydration behavior that can influence adsorption.

The full model therefore used the following ML inputs: seven operating/material numeric variables; adsorption capacity, capacity-loading ratio, and

Table 2: Feature-set definitions used to separate process-monitoring and screening tasks.

Feature set	Included information	Intended use
Full capacity process model	Operating variables, surface area, adsorption capacity, capacity-derived ratios, ion descriptors, adsorbent family, and adsorbate identity	Descriptive or process-monitoring prediction
Screening-safe plus ion	Operating variables, surface area, engineered process variables, ion descriptors, and adsorbent family; excludes capacity terms	Pre-experiment screening with ion chemistry
Screening-safe plus adsorbate one-hot	Operating variables, surface area, engineered process variables, adsorbate identity, and adsorbent family; excludes capacity terms	Screening where pollutant identity is known
Screening-safe numeric-only	Operating variables, surface area, and engineered process variables; excludes capacity and adsorbate identity	Strict screening baseline

thermo-capacity; five engineered process descriptors; six ion descriptors; normalized adsorbate identity; and broad adsorbent family. The capacity-free models retained the operating/material variables and engineered descriptors but removed adsorption capacity, capacity-loading ratio, and thermo-capacity. Among the capacity-free models, one retained ion descriptors, one used normalized adsorbate identity, and the strictest model used numeric variables only.

2.4. Machine-learning models

XGBoost was selected as the primary model because gradient-boosted trees are effective for nonlinear tabular datasets and are widely used in adsorption ML studies [10, 11]. LightGBM was included as an additional gradient-boosting comparator because preliminary model development indicated similar row-wise interpolation performance. Extremely randomized trees (ExtraTrees) were included as a comparator ensemble model because random forest-style tree ensembles provide robust nonlinear baselines for tabular prediction [12, 13]. A mean dummy regressor was included to quantify improvement beyond a no-skill baseline.

Categorical variables were one-hot encoded with unknown-category handling, while numerical variables were passed directly to the tree models. XGBoost used 140 estimators, learning rate 0.05, maximum depth 4, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 1.0. The standard LightGBM model used 300 estimators, learning rate 0.05, maximum depth 5, 31 leaves, minimum child samples 15, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 0.5. Two LightGBM sensitivity configurations were also evaluated because preliminary model development had produced strong single-split LightGBM performance. The conservative configuration used 600 estimators, learning rate 0.03, maximum depth 5, 20 leaves, minimum child samples 25, subsample 0.80, column subsample 0.80, ℓ_1 regularization 0.5, and ℓ_2 regularization 1.0. The expanded configuration used 500 estimators, learning rate 0.035, maximum depth 6, 40 leaves, minimum child samples 10, subsample 0.90, column subsample 0.90, ℓ_1 regularization 0.05, and ℓ_2 regularization 0.2. ExtraTrees used 120 estimators, maximum depth 10, and minimum leaf size 2. Predictions were clipped to the physically meaningful RE range of 0–100% before metric calculation.

2.5. Validation design

Two validation tracks were used (Fig. 2). Track A evaluated source-specific interpolation. Although the cleaned dataset contained 189 normalized source references, most references contributed only a small number of rows. Therefore, source-specific 5-fold cross-validation was reported only for references with at least 20 usable records and sufficient variation in removal efficiency, using shuffled folds with a fixed random seed. This tests whether high R^2 values are achievable within a known experimental source, which is similar to the validation setting of many high-performance adsorption ML studies.

Track B evaluated global generalization using three validation protocols within the restricted primary domain. The random row-wise split used an 80/20 train-test split stratified by the three retained pH-concentration regimes, so each split retained acidic/high-C_0, acidic/low-C_0, and neutral/high-C_0 observations. This stratification affected only how rows were assigned to training and test sets; the regime label itself was not included in the fitted model. The adsorbent-held-out split used group splitting so that exact adsorbent material names in the test set were absent from the training set; all rows for a held-out adsorbent identity were assigned to the test split. This protocol tests transfer to unseen material identities when measured descriptors such as surface area, operating conditions, pH, dose, and concentration are available. It does not claim prediction for a completely undescribed material from name alone. The reference-held-out split used group splitting so that entire references/sources were absent from the training set, making it the stricter proxy for transfer across literature sources, experimental protocols, adsorbent synthesis routes, and reporting practices. Each Track B protocol was repeated over five random seeds. Metrics included R^2, root mean squared error (RMSE), mean absolute error (MAE), and prediction bias.

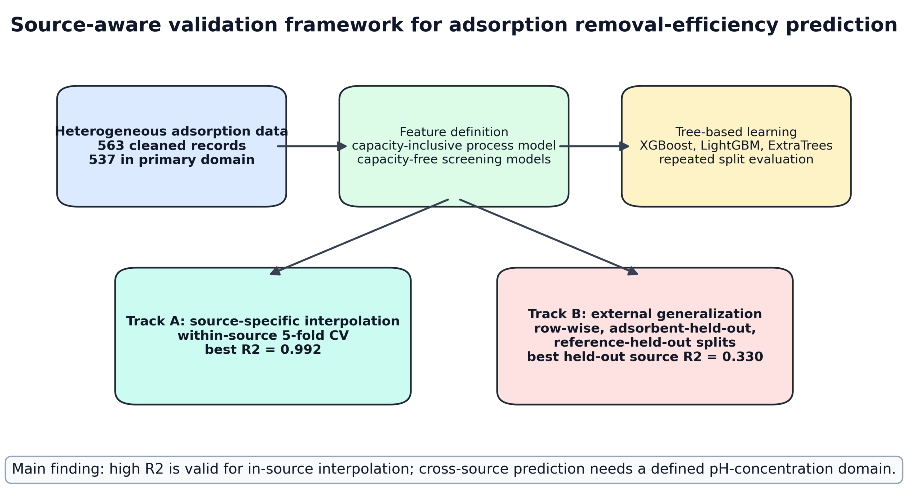

Figure 2: Workflow of the proposed source-aware adsorption-ML validation framework. Track A measures source-specific interpolation, whereas Track B measures progressively stricter global generalization.

3. Results

3.1. Source-specific models reproduced high adsorption-prediction accuracy

Track A showed that high removal-efficiency prediction accuracy is achievable when models are trained and evaluated within individual source groups. The source groups in Table 3 are not the only sources in the dataset; they are the clean bibliographic source groups that met the minimum row-count and target-variability criteria and produced stable positive within-source cross-validation performance. The full capacity process model achieved R^2 = 0.9916 for the source group Shen et al. (2017b), R^2 = 0.9735 for Gao et al. (2019), and R^2 = 0.9601 for Zama et al. (2017). These results are in the range commonly reported in adsorption ML studies under row-wise or within-domain validation [1, 3, 2]. Two additional ScienceDirect-indexed URL-bucket groups met the row-count threshold but were not treated as clean source-specific Track A groups because their reference metadata indicated mixed or unresolved bibliographic grouping. They are retained in the benchmark audit trail, but not used to support the high-interpolation claim. This reinforces the central point that source-specific predictability should be evaluated only for coherent source groups and should not be summarized by a single global accuracy value.

Capacity-free source-specific models also performed well in selected source groups. For example, the source-specific model for Zama et al. (2017) achieved R^2 = 0.9250 using the screening-safe ion feature set, while Shen et al. (2017b) achieved R^2 values above 0.91 using capacity-free screening feature sets. This indicates that high source-specific performance is not solely a capacity-inclusion artifact, although capacity-inclusive models generally gave the strongest interpolation performance.

Table 3: Track A source-specific 5-fold cross-validation results for coherent source groups with stable positive within-source predictability. Complete Track A benchmark outputs, including unresolved ScienceDirect URL-bucket groups, are retained in the submitted CSV files for auditability.

Source group	Rows	R^2	RMSE	MAE
Shen et al. (2017b)	65	0.9916	3.20	2.33
Gao et al. (2019)	48	0.9735	2.97	2.14
Zama et al. (2017)	91	0.9601	7.84	3.56
Cui et al. (2016a)	24	0.8936	6.65	4.56

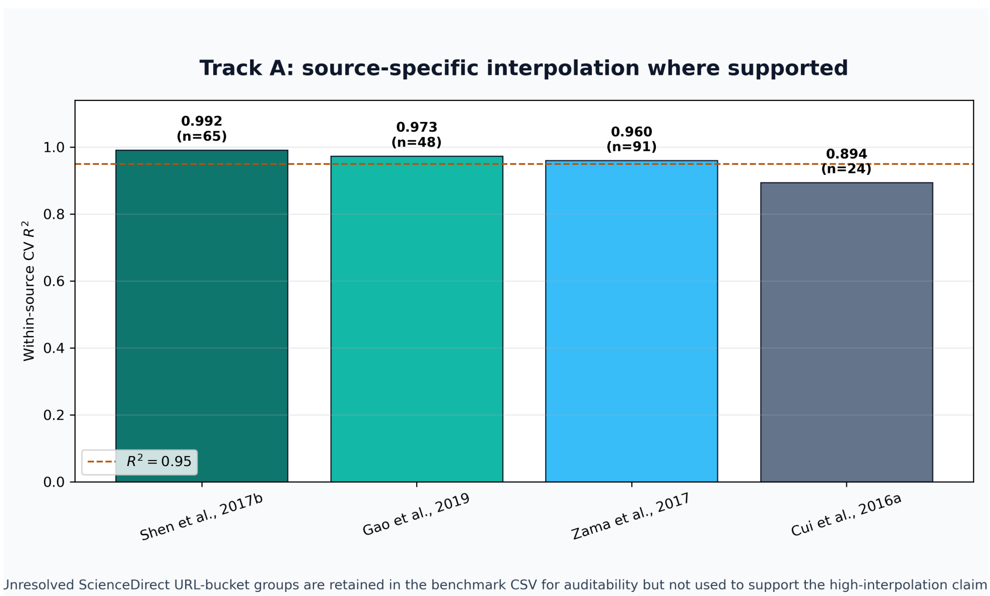

Figure 3: Track A source-specific interpolation performance. Several individual source-group models achieved R^2 values near or above 0.95, while other source groups remained more difficult.

3.2. Random row-wise validation produced moderate-to-good global performance

Under repeated random row-wise splits in the restricted primary domain, the full capacity process model achieved R^2 = 0.8064 and RMSE = 12.94 with LightGBM, while the corresponding XGBoost model achieved R^2 = 0.7985 and RMSE = 13.22. The screening-safe plus adsorbate one-hot XGBoost model achieved R^2 = 0.7283 and RMSE = 15.29, while the screening-safe plus ion XGBoost model achieved R^2 = 0.7265 and RMSE = 15.34. Thus, even without adsorption capacity, the model retained useful predictive ability under in-dataset interpolation (Table 4).

Table 4: Restricted-domain random row/regime split performance after excluding the sparse neutral/low-C_0 regime. Values are means over five repeated splits.

Feature/model	R^2	RMSE	MAE
Full capacity LightGBM	0.8064	12.94	8.08
Full capacity XGBoost	0.7985	13.22	8.44
Full capacity ExtraTrees	0.7730	14.02	9.02
Screening-safe plus adsorbate XGBoost	0.7283	15.29	10.84
Screening-safe plus ion XGBoost	0.7265	15.34	10.88

These results support the use of ML for in-domain adsorption prediction. They also explain the strong preliminary LightGBM results obtained during model development. Earlier LightGBM reports used a single random/regime-stratified split and included capacity-related predictors, giving test R^2 = 0.847 and RMSE = 12.05; in the present repeated-split benchmark, individual LightGBM random splits similarly reached R^2 = 0.8499 for the standard configuration and

R^2 = 0.8570 for the conservative sensitivity configuration. However, the five-split means were lower (R^2 = 0.8064 and R^2 = 0.7983, respectively), showing that the earlier value was a valid favorable row-wise interpolation result rather than a stable estimate of source-held-out generalization. Random row-wise validation alone is therefore not sufficient evidence for deployment on unseen sources.

The two capacity-free XGBoost screening models differed in how the adsorbate was represented. The screening-safe plus adsorbate model used normalized adsorbate identity as a categorical one-hot feature, allowing the model to learn pollutant-specific effects directly. The screening-safe plus ion model used simple aqueous-ion descriptors, including charge, ionic radius, hydrated radius, ionic potential, and hydration ratio, instead of the direct pollutant label. The similar row-wise performances of these two models, R^2 = 0.7283 and R^2 = 0.7265, respectively, indicate that basic ion-chemistry descriptors captured much of the information provided by explicit adsorbate labels in the random-split setting.

3.3. Cross-material and cross-source validation tested broader transfer

The validation protocol strongly affected performance, which is expected for a heterogeneous dataset spanning many source references, adsorbent identities, and metal ions. In the adsorbent-held-out protocol, all experiments associated with selected adsorbent material names were removed from the training set and used for testing. This is different from holding out adsorbates: the held-out groups are adsorbent materials, not pollutant ions. Under this cross-material test, the best capacity-free result was obtained by the screening-safe numeric expanded-LightGBM model, with R^2 = 0.5323 and RMSE = 19.35. The screening-safe numeric XGBoost model achieved R^2 = 0.5199, while the conservative full-capacity LightGBM model achieved R^2 = 0.5258. These results indicate moderate but imperfect transfer to unseen adsorbent identities when measured descriptors and operating conditions are available.

Reference-held-out validation was the most important broad-source stress test because every row from selected literature references was kept outside training. The model was therefore evaluated on source domains whose experimental protocol, adsorbent preparation, concentration window, pollutant distribution, and reporting pattern were not represented by the same reference during training. This validation is closer to the practical question of whether a literature-trained model transfers to a new study. It was more difficult than row-wise validation but improved markedly after removing the sparse neutral/low-C_0 regime. The best reference-held-out sensitivity result was obtained by the full-capacity expanded-LightGBM model, with $R^2 = 0.3296$ and RMSE = 22.88. The best capacity-free reference-held-out result remained the screening-safe numeric-only ExtraTrees model, with $R^2 = 0.3054$ and RMSE = 23.54. These results remain below row-wise performance, but they show that defining a better-covered applicability domain produces more meaningful external-source validation for a broad, multi-source adsorption dataset.

These results show that removal-efficiency prediction is strongly source-dependent. The same data and model family can produce high R^2 under source-specific interpolation, moderate-to-good R^2 under random global validation, and lower but still informative R^2 under source-held-out validation when the applicability domain is restricted to sufficiently represented pH-concentration regimes.

Table 5: Restricted-domain Track B generalization performance after excluding the sparse neutral/low-C_0 regime. Values are means over five repeated splits.

Validation	Feature/model	R^2	RMSE	MAE
Holdout adsorbent	Screening-safe numeric expanded LightGBM	0.5323	19.35	14.55
Holdout adsorbent	Full capacity conservative LightGBM	0.5258	19.19	13.00
Holdout adsorbent	Screening-safe numeric-only XGBoost	0.5199	19.61	15.15
Holdout reference/source	Full capacity expanded LightGBM	0.3296	22.88	17.67
Holdout reference/source	Screening-safe numeric-only ExtraTrees	0.3054	23.54	18.67
Holdout reference/source	Full capacity standard LightGBM	0.2820	23.78	18.67

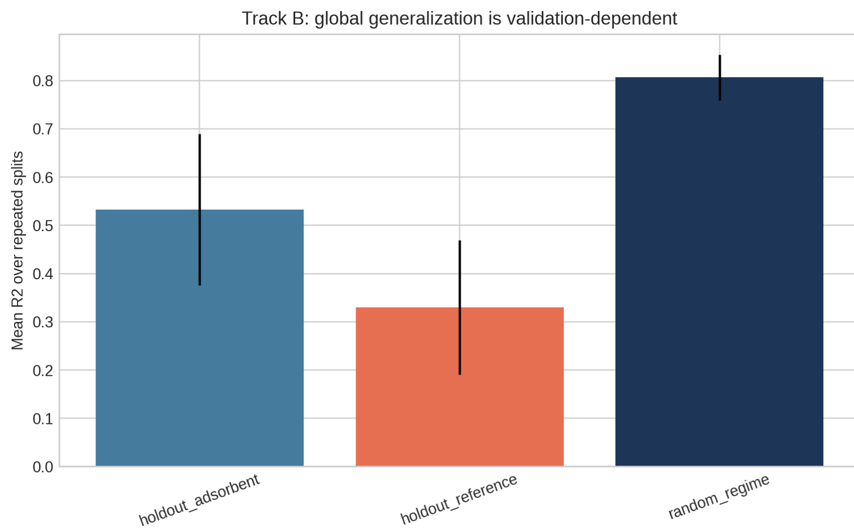

Figure 4: Track B validation comparison. Row-wise validation, adsorbent-held-out validation, and reference-held-out validation represent increasingly strict generalization tasks.

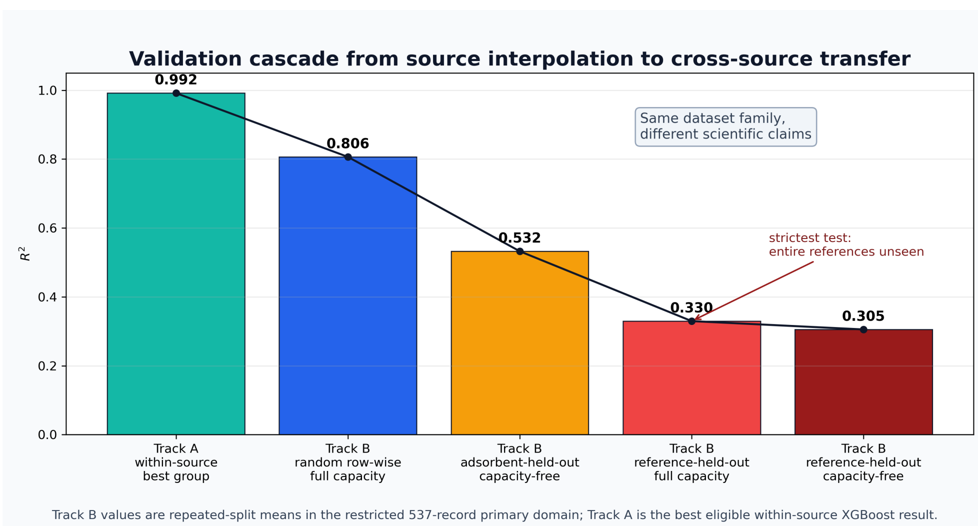

Figure 5: Validation cascade from source-specific interpolation to stricter global transfer. The figure summarizes why high R^2 values near 0.95 are achievable within known sources while reference-held-out prediction is a more difficult external-source task.

3.4. Capacity-inclusive and capacity-free models answered different questions

The full capacity process model performed best for random row-wise validation and several source-specific models. This is expected because adsorption capacity carries strong information about adsorption performance. However, capacity can be outcome-adjacent to RE and may not be available before an experiment. Therefore, capacity-inclusive models should be framed as process-monitoring or descriptive models. Capacity-free models are more appropriate for pre-experiment screening and adsorbent-design guidance. In the restricted-domain adsorbent-held-out and reference-held-out tests, capacity-free numeric models were competitive or superior, suggesting that simpler descriptor sets can be more robust when transferring across material identities or source studies.

4. Discussion

4.1. Why near-0.95 R^2 is achievable but validation-dependent

The Track A results explain why many adsorption ML studies report R^2 values near or above 0.95. Within a single source, adsorbent family, pollutant system, or experimental protocol, the response surface can be learned accurately by tree-based models. In this setting, high R^2 is scientifically plausible and not automatically invalid. The present results reproduce that behavior, with source-specific R^2 values of 0.960–0.992 for three major source groups.

However, Track B shows that this performance should not be interpreted as universal adsorption prediction. When sources are held out, the model must transfer across experimental protocols, adsorbent synthesis methods, pollutant distributions, concentration ranges, and reporting conventions. Under that validation setting, performance fell below row-wise performance even after excluding the sparse neutral/low-C_0 regime. The result is consistent with recent adsorption-related work showing that random-split performance can be substantially higher than leave-compound-out performance [9].

4.2. Relation to existing adsorption ML literature

The proposed framework complements rather than contradicts previous high-accuracy adsorption ML studies. Hafsa et al. demonstrated that tree-based algorithms can achieve high removal-efficiency prediction accuracy for heavy-metal adsorption datasets under repeated cross-validation [1]. Wang et al. showed that XGBoost, SHAP, and partial-dependence analyses can identify interpretable drivers of heavy-metal removal by biochar [3]. Lu et al. showed that random forests can predict heavy-metal removal by chitosan-based flocculants with R^2 above 0.93 [2]. Barkhordari et al. further demonstrated that interpretable XGBoost models are relevant to JECE's environmental-remediation scope by predicting heavy-metal removal efficiency in electrokinetic remediation [5]. These studies establish that ML is useful for environmental treatment modeling.

The present study adds a validation-layer contribution: adsorption ML workflows should explicitly state whether they are predicting within a known source or generalizing to unseen sources. This distinction is critical because high row-wise R^2 can be overoptimistic when the deployment target is a new laboratory, adsorbent synthesis route, pollutant matrix, or literature source. The broader ML literature has identified leakage and validation mismatch as major reproducibility risks [7, 8], and adsorption datasets have several mechanisms by which such mismatch can occur.

4.3. Practical implications for adsorption model deployment

For practical adsorption screening, a single universal RE model should be used cautiously. A more defensible workflow is to define the pH-concentration applicability domain before modeling, exclude or separately model poorly represented regimes, use source-specific or laboratory-calibrated models for high-accuracy optimization within a known experimental system, use capacity-free screening models for preliminary adsorbent and operating-condition selection, and report source-held-out validation when predictions are intended for new adsorbents, pollutants, laboratories, or literature domains. This framing allows high-R^2 models to remain useful without overstating their generality.

4.4. Limitations and future work

This study has four main limitations. First, the dataset does not contain explicit laboratory-origin labels for all records; bibliographic reference was therefore used as a source proxy. Future work should annotate records as in-house, external-laboratory, or literature-derived and repeat validation with leave-laboratory-out splits. Second, the ion descriptor table should be expanded with fully cited descriptors such as electronegativity, hydration energy, softness/hardness, and likely aqueous speciation. Third, important adsorbent descriptors such as pH_{pzc}, pore volume, pore size, zeta potential, FTIR-derived functional groups, and cation-exchange capacity are incomplete in the present dataset. Fourth, the neutral/low-concentration regime contained only 26 observations and was excluded from primary modeling; predictions in that regime should therefore be treated as outside the present applicability domain until more data are available. These limitations define clear routes for improving external predictive performance while preserving the source-aware validation principle.

5. Conclusions

This study developed a source-aware and leakage-aware framework for adsorption removal-efficiency prediction. High R^2 values were achievable within individual adsorption sources, with source-specific XGBoost models reaching R^2 up to 0.992. Because the neutral/low-concentration regime contained only 26 observations, primary global modeling excluded that sparse regime and used a 537-record restricted domain. Within this domain, random row-wise validation reached $R^2 = 0.806$ for the full capacity LightGBM process model, adsorbent-held-out validation reached $R^2 = 0.532$ for the capacity-free numeric expanded-LightGBM screening model, and reference-held-out validation reached $R^2 = 0.330$ for the full-capacity expanded-LightGBM sensitivity model. The best capacity-free reference-held-out model reached $R^2 = 0.305$. These findings show that high-accuracy interpolation, applicability-domain definition, and cross-source generalization are different scientific tasks. Future adsorption ML studies should report all three, separate capacity-inclusive process-monitoring models from capacity-free screening models, and include source- or laboratory-held-out validation when deployment beyond the training source is claimed.

CRediT authorship contribution statement

Yash Bisht: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. Susmit Chitransh: Data curation, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The curated dataset, analysis scripts, and generated benchmark outputs are available from the corresponding author upon reasonable request and will be deposited in a public repository before final publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's Codex to assist with code organization, LaTeX formatting, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

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